ALLOC-X

PRODUCTION ARCHITECTURE

AI proposes. Controls decide. Signers authorize.

XRPL settles. Independent monitoring proves the result.

Autonomous • protected keys

APPLICATION DOMAIN

Human interaction, integrations and agent planning

1 REQUEST AND EVIDENCE

Console
Finance and treasury operators

Systems
ERP, procurement and payroll

Offline field
Encrypted drafts; sync before payment

2 AGENTIC PLANNING

Allocation Agent
Plans and initiates the complete autonomous payment journey.

Planning Core
Resolves authoritative balances, obligations, schedules and identifiers.

AUTHORITATIVE SOURCES

- Policy registry
- Beneficiaries
- Evidence
- Treasury state
- Obligations
- Accounting

Documents are evidence—not executable instructions

typed draft

A

INTENT BOUNDARY

Typed, canonical and replay-safe

3 CANONICAL INTENT

PaymentIntent
Schema-validated request bound to authoritative IDs and a stable digest.

- Fixed amount
- Beneficiary version

BOUND EXECUTION

Amount, asset, allocation, destination version, evidence and policy are all hashed together.

sha256: exact intent

NON-NEGOTIABLE

- No free-text wallet
- No browser-supplied amount
- Every retry is idempotent
- Any mutation creates a new digest

evaluate

B

INDEPENDENT SECURITY DOMAIN

Approval Layer must sign before money can move

4 APPROVAL LAYER

Treasury Security Gateway
Fails closed • independently authenticated

SIGNED RESULT

short-lived

1	Identity & authority	CHECK
2	Policy & allocation	CHECK
3	Beneficiary integrity	CHECK
4	Evidence integrity	CHECK
5	Fraud & anomaly	CHECK
6	Treasury safety	CHECK
7	Ledger semantics	CHECK
8	Replay & velocity	CHECK
9	Final equivalence	CHECK

ALLOW · REVIEW · BLOCK

REVIEW / BLOCK returns reasons and safe next actions

ALLOW digest

C

SETTLEMENT DOMAIN

Exact transaction and signer quorum

5 AUTHORIZE AND SETTLE

Payment Orchestrator
Builds the exact XRPL transaction and submits exactly once.

D

Independent Signers
Execution signer + policy signer. Each recomputes the transaction digest.

No submit access

E

XRPL Adapter
Fee, sequence, serialization, multisign assembly and submission.

submit

F

AUTONOMOUS AUTHORITY

Allowed transactions sign and settle automatically with protected keys.

PUBLIC LEDGER

XRP Ledger
Validated settlement

FINAL

ON-LEDGER

- Tx hash
- Ledger index
- Receipt anchor

OFF-LEDGER

Encrypted evidence and full receipt

independent validated-event feed

6 INDEPENDENT MONITORING, PROOF AND RECOVERY

A separate read-only path observes XRPL; the submitter cannot declare itself successful.

Ledger Monitor
Read-only XRPL observer accepts only validated successful payments.

Reconciliation
Matches expected settlement and finalizes accounting reservations.

Decision Receipt
Portable proof of intent, policy, evidence, signatures and ledger result.

Audit Chain
Append-only hash chain detects deletion, mutation or reordering.

Operations & Recovery
Alerts, emergency freeze, key rotation, retry recovery and unknown-outflow response.

TRUST MODEL — Application control · Independent verification · Signing authority · Human review · Blocked / recovery

Alloc-X • production target architecture